Deep Learning for Enhanced Ocean Color Remote Sensing: A Foundation Model Approach

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***Abstract*—Ocean color remote sensing has advanced over the past 40 years, with algorithms primarily rooted in radiative transfer theory to estimate various ocean color properties. To address the diversity and region-specific distribution of these properties, numerous algorithms have been developed; however, each is tailored to specific variables. In recent years, deep learning has demonstrated remarkable progress in this field, but a general foundational model for robust ocean color product generation remains lacking. To address this limitation, this study introduces the ocean color deep-learning foundation model (OCFM), a comprehensive framework designed to extract multiple global ocean color properties from satellite observations. The development of OCFM follows three sequential phases: pretraining with operational satellite products to learn fundamental theoretical relationships, fine-tuning with in situ measurements to align with the real ocean state, and deploying the trained OCFM for flexible retrieval of ocean color properties. Quantitative evaluation shows that, in end-user application tests, the model achieved a coefficient of determination (R^2) of 0.90 for primary productivity and 0.71 for water clarity, with corresponding mean absolute percentage differences of 44.08% and 17.28%, respectively. OCFM provides a more efficient solution for few-shot and unevenly distributed samples compared with traditional retrieval algorithms. Even with limited user resources and small sample sizes, additional downstream training with the trained OCFM can achieve state-of-the-art performance in retrieving ocean color properties. This study highlights the potential of a deep-learning foundation model for generalizable and few-shot ocean color retrieval.**

***Index Terms*—Deep learning, foundation model, multitask learning, ocean color, remote sensing.**

I. Introduction

OCEAN color measurement and analysis are crucial for Earth observation, significantly contributing to water quality monitoring, ecological studies, and fisheries management. Ocean color remote sensing is essential for obtaining large-scale spatial and temporal information on marine biogeochemical processes [1], [2]. The primary objective of ocean color remote sensing is to achieve high-precision apparent optical properties (AOPs) from satellite imagery and subsequently derive the inherent optical properties (IOPs) of the water body along with the concentrations of associated substances [3], [4]. AOPs, including remote sensing reflectance (R_{rs}) and downwelling irradiance (E_d), which are influenced by the geometry of the radiation distribution. IOPs include absorption and backscattering coefficients, which are independent of the ambient light field, depend solely on the water and its constituents, and are crucial for understanding and interpreting ocean environments.

The R_{rs} spectrum of AOPs is fundamental for deriving IOPs, and all satellite ocean color algorithms rely directly or indirectly on R_{rs} [5], [6], [7]. Based on satellite-derived R_{rs}, numerous retrieval models have been developed to estimate various global IOPs, such as suspended particulate matter (SPM) [8], [9], colored dissolved organic matter [10], [11], [12], water clarity [13], [14], and phytoplankton pigments [15], among others [16], [17], [18]. Other regional algorithms are also designed for specific regions with distinct geographic characteristics [19], [20], [21].

These algorithms can be broadly classified into three main types, each with distinct methodological foundations and application characteristics. Semi-analytical algorithms are based on radiative transfer theory and seek to establish mechanistic relationships between R_{rs} and IOPs through the inversion of forward models [6], [8], [9], [10], [14], [19], [20], [21]. Due to their basis in physical principles, these algorithms exhibit relatively consistent formulations across different water types and offer good interpretability and theoretical rigor. Empirical algorithms rely on statistical relationships derived from field observations and are typically tailored to specific regions or water types [11], [12]. Compared with semi-analytical algorithms, they are easier to implement and more computationally efficient, making them suitable for rapid applications in well-characterized environments. Supervised learning encompasses algorithmic models such as machine learning and deep learning, which rely on iterative optimization of parameters to learn complex patterns from data [16], [17], [22]. These models offer greater flexibility and predictive performance, particularly in data-rich environments with diverse and representative training samples.

Despite their utility, existing ocean color algorithms face several limitations that hinder their scalability and generalization.

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1) Most existing ocean color algorithms are designed to retrieve only a single variable or a limited set of closely related IOPs. However, radiative transfer theory indicates that the top-of-atmosphere signal is inherently a mixture of contributions from various water constituents [22], [23]. This interdependence implies that information obtained from one component can provide valuable constraints or contextual support for estimating others. Algorithms that isolate the retrieval of individual parameters often overlook these cross-component relationships, leading to suboptimal use of the information embedded in the R_{rs} spectrum. As a result, such models may suffer from reduced accuracy or increased uncertainty when applied in optically complex or dynamic aquatic environments [17].
2) The development of most existing algorithms is task-specific and lacks flexibility for adaptation to emerging variables or new retrieval objectives. When novel biogeochemical or ecological parameters become of interest, such as new water quality indicators or ecosystem health, existing models often cannot be directly repurposed. Semi-analytical approaches require new forward-inversion formulations; empirical algorithms demand new calibration datasets; and supervised learning models must be retrained from scratch with annotated examples. This lack of transferability greatly limits the scalability of traditional methods and delays the response to evolving scientific and operational needs in ocean color remote sensing. Moreover, supervised learning models are equally dependent on representative and sufficiently large training datasets, which further limits their generalizability across diverse environments.
3) Most models, including semi-analytical, empirical, and learning-based approaches, require iterative updates of parameters or retraining to adapt to changes in sensor characteristics, environmental conditions, or regional optical properties [6]. For example, the performance of analytical algorithms heavily depends on accurate parameter tuning and assumptions about the IOPs of specific water types. These dependencies can lead to suboptimal retrievals when applied outside of the conditions for which they were originally calibrated [24].

SeaDAS integrates various semi-analytical algorithms for atmospheric and oceanic product generation. For instance, the generalized ocean color inversion model (GIOP) framework [25] allows users to assemble inversion algorithms from a configurable library of shape functions, facilitating standardized comparisons and community alignment. While this represents a significant advancement, SeaDAS remains a collection of modular, rule-based algorithms. Each use case typically requires manual parameter tuning and prior knowledge, limiting automation and flexibility. Moreover, its architecture does not support gradient-based optimization, making it incompatible with modern deep-learning paradigms.

Recent advances in deep-learning foundation models offer a new paradigm that can overcome these limitations [26], [27], [28]. A foundation model is a large neural network trained on broad data and adapted to many downstream tasks [29].

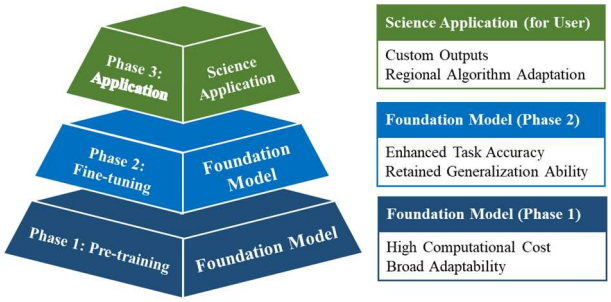


Fig. 1. Multiphase workflow of the foundation model within the ocean color retrieval domain.

Such models shift away from single-purpose training toward a two-phase approach: first, pretraining on massive datasets to learn general features, then lightweight fine-tuning or modular extension for each specific task [29], [30]. Foundation models enable generalization to new tasks with minimal labeled data by capturing shared structure through large-scale pretraining. They support modular design, allowing new tasks to be added with minimal retraining. This architecture addresses previous limitations by replacing redundant, task-specific models with a unified architecture that jointly retrieves multiple ocean color properties. Since the model encodes common radiative transfer variability, only limited task-specific fine-tuning is required. Retrieval algorithm development is thus streamlined into lightweight task-specific modules built on a shared backbone.

We redefine the foundation model paradigm for ocean color retrieval through a structured, three-phase workflow (see Fig. 1), replacing monolithic pipelines with a modular architecture that separates representation learning from task specialization.

Phase 1 involves large-scale pretraining on billions of satellite-derived algorithmic products to learn generalizable spectral–spatial representations. Despite its high computational cost, this phase yields broad adaptability across oceanic conditions.

Phase 2 fine-tunes the pretrained model using a few hundred thousand high-quality in situ measurements to improve task-specific accuracy while retaining the generalization capacity of the backbone.

Phase 3 delivers an application-ready model for end-users, allowing rapid deployment of custom outputs (e.g., primary productivity and water clarity) and regional algorithm adaptations through lightweight plug-in modules with users' own data, without retraining the full network.

As illustrated in Fig. 1, each phase builds upon the last: pretraining provides a broad foundation, fine-tuning sharpens task precision, and application enables operational flexibility. This sequential design avoids redundant modeling and supports scalable extensions. This architecture reduces data requirements by over 99% compared with conventional multitask models, enables fast fine-tuning on standard hardware, and streamlines retrieval algorithm development by assembling compact task-specific heads on a unified pretrained backbone.

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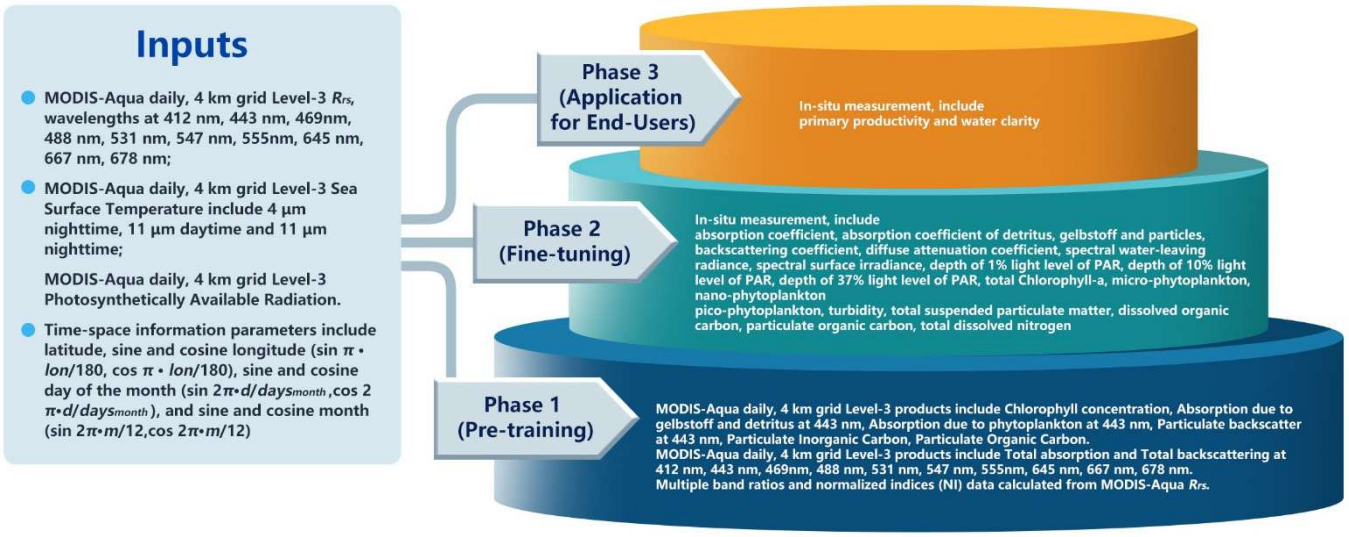


Fig. 2. Three-phase dataset of inputs and target of output variables.

This article's contributions are: 1) introducing the OCFM architecture as a foundation model approach for ocean color inversion and 2) showing how multitask learning within this framework reduces redundancy and harnesses cross-variable relationships. The remainder of this article is organized as follows. Section II describes the datasets and preprocessing. Section III details the OCFM model design. Section IV presents experiments and results. Finally, Section VI concludes with a discussion and future directions.

II. Dataset

Considering that MODIS-Aqua provides a significantly longer time series of observations compared with other ocean-color satellites and aligns with richer, more temporally extensive in situ data, we selected MODIS-Aqua as the primary satellite data source for this study. In addition, interfaces for integrating data from other satellites have been retained to ensure future extensibility. Future data from other satellites can also be harmonized into the MODIS-Aqua data format through cross-satellite atmospheric correction [31].

The entire dataset consists of a pretraining set and a fine-tuning set with four sources: 1) MODIS-Aqua AOP data (e.g., R_{rs} from satellite observations); 2) MODIS-Aqua products derived from various algorithms; 3) multiple band ratios and normalized indices (NIs) data calculated from MODIS-Aqua R_{rs}; and 4) ocean color in situ measurement data. The model inputs (predictors) for the pretraining and fine-tuning processes are the MODIS-Aqua AOP data. The pretraining process uses MODIS-Aqua products, band ratios, and NI data as the target of the pretraining dataset. The training is then fine-tuned with in situ measurement data as the target of the training dataset. The input and output variables at each phase as shown in Fig. 2.

A. Satellite Data and Products

Satellite measurements from daily 4-km level-3 MODIS-Aqua are used as model predictors, comprising level-3 R_{rs} (λ = 412, 443, 469, 488, 531, 547, 555, 645, 667, and 678 nm), sea surface temperature (SST) (4-μm nighttime, 11-μm daytime, and 11-μm nighttime) and photosynthetically active radiation (PAR). MODIS-Aqua satellite data from the MODIS instrument on the NASA Aqua satellite are part of the Earth Observing System (EOS) to monitor global climate and environmental changes. MODIS has 36 spectral bands, with ten bands (412–678 nm) primarily used for ocean color, phytoplankton pigments, and biogeochemistry. The R_{rs} of these ten bands are therefore used as inputs to the model. In addition, SST and PAR, which significantly influence phytoplankton pigments and biogeochemical processes, are also included as inputs [32], [33], [34]. The MODIS-Aqua level-3 data used in this study are globally gridded and cover the period from July 4, 2002, to December 31, 2023.

Multiple ocean color products, produced and validated by the NASA Ocean Biology Processing Group (OBPG), are collected as the targets of the pretraining dataset of the pretraining set [35]. Those ocean color products include satellite chlorophyll concentration, absorption, and backscatter coefficients, K_d at 490 nm, Particulate inorganic carbon, and Particulate organic carbon (absorption, backscatter coefficients with ten bands ranging from 412 to 678 nm). These targets, derived from various operational satellite ocean color algorithms, including a blend of the updated OC3/OC4 (OCx) algorithm [36] and the color index (CI) [37] for chlorophyll-a, as well as GIOP for estimating IOPs [25], among others [38], [39], [40], with inherent accuracy limitations, are utilized to initialize the model (pretraining). Using existing products as targets ensures that the network learns the fundamental theoretical relationships for ocean color inversion during the initial pretraining phase.

B. In Situ Data

Ocean color in situ measurement data consists of published in situ measurements gathered from various databases and studies. These data are largely classified based on the basic

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optical properties of the water column (absorption coefficients, attenuation coefficients, etc.) and environmental monitoring factors (turbidity, phytoplankton pigments, etc.). Table S1 of the Supplementary Material lists the collected specific types and sources of in situ measurements. These datasets are derived from publicly available data from SeaBASS [41], [42], NASA bio-optical marine algorithm dataset (NOMAD) [43], [44], marine ecosystem data (MAREDAT) [45], global ocean data analysis project, version 2 (GLODAPv2) [46], and others. The in situ data, covering various depths and over 6.5 million points since July 1972, were filtered to exclude pre-July 4, 2002 data (11% of the total) for MODIS alignment, and the top and bottom 0.5% of extreme values were removed. Due to the variations in values and the stability of data sampled at different depths, we retained only the data for samples collected above the 10-m water column. Data for the same variable, sampled on the same day and within 4 km (within the same pixel of satellite data), were combined and averaged. After the above data screening and initial quality control, only about 3.4%, about 220 thousand points, of the total collected in situ data were subject to subsequent matchups. These quality-controlled and screened in situ measurement data will be used as targets of a training dataset for fine-tuning the model. This ensures the OCFM to learn a broader range of features and achieves more accurate IOP estimates during the fine-tuning process.

Within this in situ dataset, phytoplankton size classes (PSCs) data were obtained from high-performance liquid chromatography (HPLC) in situ data by diagnostic pigment analysis (DPA) calculation [47], [48], [49]. PSCs are closely related to marine ecosystems. PSCs were calculated from HPLC phytoplankton pigments using the DPA method [48], [50], with the updated parameters from Brewin et al.'s [51] method. For phytoplankton pigment measurements, total chlorophyll-a (TChla) samples smaller than the detection line (0.001 mg·m^{-3}), which accounted for <0.1% of the TChla, were removed [33]. In this study, the term "chlorophyll-a" follows the conventions of the respective data sources. For satellite-derived products from NASA OBPG, it refers to the TChla concentration estimated from ocean color algorithms (e.g., OCx), including all isomers and epimers contributing to the water-leaving radiance. For in situ measurements, we adopt the SeaBASS definition of TChla derived from HPLC analysis. Although the naming conventions differ, both definitions represent the TChla pigment concentration, and their equivalence is widely recognized in the ocean color remote sensing community [33], [51].

C. Matchups and Model Training Database

The inputs and corresponding output targets for both pretraining and fine-tuning must be paired one-to-one. In theory, various ocean color properties should exhibit stronger correlations with satellite R_{rs} data sampled on the same day. However, single-day observations may provide insufficient information due to gaps in satellite measurements, cloud cover, and a lack of temporal and spatial details. Temporal responses over short periods are more prominent and exhibit stronger correlations than spatial effects [52], a finding also confirmed

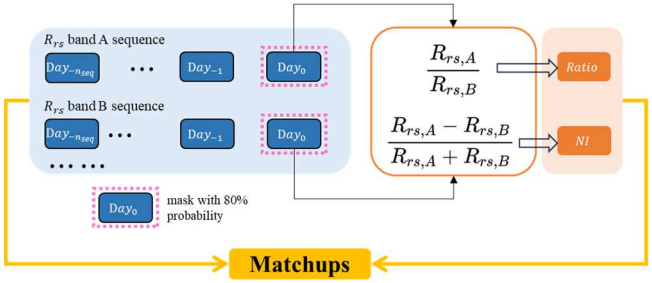


Fig. 3. Matching process of pretraining Ratio and NI database. To ensure the model learns the Ratio and NI patterns from the sequence, rather than relying on day 0 data, the input sequences for the last day are randomly masked by setting them to 0.

by experiments in Section S2 of the Supplementary Material. Consequently, we use time-series data of fixed sequence lengths as model inputs. Identifying the optimal sequence length poses a challenge. To tackle this, we conducted tests on the retrieval of TChla concentrations using remote sensing data from n days prior. Accordingly, each target corresponds to a satellite predictor time series spanning n_{seq} days.

To identify the most relevant temporal sequence n_{seq} for modeling, we analyzed the relationship between satellite observations from the preceding n days and the current-day TChla concentration. Since this relationship may be nonlinear, we conducted a series of prediction experiments using five machine learning models: artificial neural network (ANN), random forest regression (RFR), gradient boosting regressor (GBR), support vector regression (SVR), and decision tree regressor (DTR). While prediction was not the primary objective, the resulting prediction accuracy served as a proxy to evaluate the strength of correlation across different values of n_{seq}. These experiments indicated that using satellite data from the previous 30 days yielded the best performance, and we therefore selected $n_{\text{seq}} = 30$ as input length. Further details of this analysis are provided in Section S2 of the Supplementary Material.

As a common practice for deep-learning models, performance enhancement typically involves expanding the pretraining set to initialize weights and capture underlying data characteristics [53], [54]. This study's targets of pretraining dataset include satellite products, band ratios, and NI extracted from individual pixel points. Ratios and normalization variables are calculated for any two R_{rs} bands, as shown in Fig. 3. Adding this information helped diversify the tasks, as R_{rs} band ratios and NI are closely linked to various ocean optical properties, and many empirical regression models rely on these ratios [12], [55], [56]. For a given sequence of R_{rs} extracted from satellite imagery, band ratios and NI are computed using only the R_{rs} values from the last day in the sequence. These computed values serve as a target label for phase 1, aiming to guide the model in capturing meaningful sequence-level features. However, since the R_{rs} data from the final day are used both as part of the input and for label generation; there is a risk of information leakage—where the model could directly learn to reconstruct the target from the corresponding input at that timestep. To mitigate this, we apply a masking strategy

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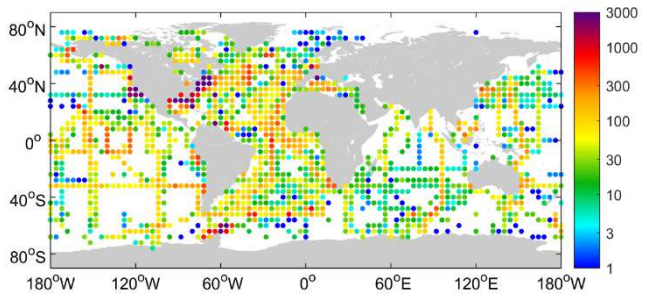


Fig. 4. Global distribution of in situ measurement points used in the fine-tuning dataset, with color indicating the number of samples collected.

during training: specifically, 80% of the R_{rs} values from the last day are randomly masked in the input sequence. This approach forces the model to rely on contextual information from previous days rather than directly referencing the final-day input, thereby simulating realistic missing data conditions and improving generalization. The 80% masking probability was selected to closely align with the typical missing data rate observed in real-world satellite R_{rs} products, which generally range from 70% to 80%. This setting improves the model's robustness to both intentionally masked inputs during training and naturally missing data during inference.

A total of 7377 days of global satellite observations, from July 2002 to September 2023, were used to construct the pretraining dataset, which includes AOPs such as remote sensing reflectance (R_{rs}); satellite-derived geophysical products generated through algorithmic processing of PAR; and SST. Each valid pixel in the daily global images was treated as an independent training sample, resulting in over 3.8×10^{10} samples for large-scale pretraining of OCFM.

In the in situ dataset, the training database comprises matchups between all in situ measurements and corresponding sequences of satellite predictors. MODIS-Aqua level-3 data do not include explicit quality assurance flags. Initially, we masked only "Not a Number" (NaN) values. To further exclude potentially low-quality R_{rs} pixels, we followed the approach of Bailey and Werdell [43] by calculating the coefficient of variation (CV) within a 3×3 pixel window. The CV was defined as the ratio of the filtered standard deviation to the filtered mean for $R_{rs}(\lambda)$ across wavelengths from 412 to 555 nm, as well as for aerosol optical thickness at 865 nm. Pixels with a median CV exceeding 0.15 were excluded from training [33]. These discarded pixels were treated as zero-incentive values in the model input. The global distribution of in situ measurements is shown in Fig. 4. Ultimately, we obtained a total of 105 440 in situ matchups.

III. Methods

A. Adapted OCFM Architecture for Ocean Color Retrievals

In developing the OCFM, we implemented three phases, as shown in Fig. 5.

1) *Pretraining:* The model was initialized using widely accepted ocean color products as pretraining dataset targets, enabling it to learn fundamental ocean color algorithms.
2) *Fine-Tuning:* The model was further refined with a large set of in situ measurements, improving its accuracy.
3) *Application:* End-users can train the model downstream submodules with their datasets, allowing it to adapt to specific tasks. For less available ocean color data types, the retrieval relies not only features of the smaller dataset itself but also shared correlations and interproperties relationships derived from ocean color products and diverse in situ measurements.

Furthermore, two operations were implemented for OCFM to facilitate ocean color retrieval tasks. First, we utilized time-series data with a defined sequence length as model inputs, which improved the input data characteristics. Temporal responses of marine environments over short periods are stronger and more correlated than spatial effects [52], [57], [58], [59]. Second, the diversity of training data was expanded. During the pretraining phase, besides using standard ocean color products as targets, we also employed band ratios and the NI of R_{rs} as additional targets. In the fine-tuning phase, in situ measurements included data on phytoplankton, chromophoric dissolved organic matter (CDOM), and SPM, covering all IOP types. Additional water quality monitoring factors and AOPs further enriched the fine-tuning dataset.

B. OCFM Framework Design

The OCFM architecture consists of a multihead attention (MHA) [60] neural network for effectively capturing disparate information in input data and Fourier neural operators (FNOs) [61] to enhance the model's computational power. OCFM framework is divided into three parts: an inEmbed for encoding satellite observation, an encoder–decoder structure as the backbone to unify and process the shared learning parameters, and multiple expandable outEmbeds for different subtasks, as shown in Fig. 6. The inEmbed structure and encoder–decoder structure contain sufficient learnable parameters to ensure the overall learning performance of the neural network. In contrast, the outEmbeds do not need to be particularly large; they only need to transform the backbone network learned features (denoted as "output data embeddings") into their corresponding ocean color variables.

1) InEmbed: The inEmbed accepts input from satellite observation sequences and other time-space information parameters and then encodes the input data after FNO to the encoder–decoder backbone. Other time-space information parameters include latitude, sine and cosine longitude $(\sin \pi \cdot \text{lon}/180, \cos \pi \cdot \text{lon}/180)$, sine and cosine day of the month $(\sin 2\pi \cdot d/\text{days}_{\text{month}}, \cos 2\pi \cdot d/\text{days}_{\text{month}})$, and sine and cosine month $(\sin 2\pi \cdot m/12, \cos 2\pi \cdot m/12)$. By training the different inEmbed module, the model can be expanded to incorporate data from various satellites in the future.

The satellite sequence observations for the time-series input with shape $\text{seq} \times C_1$, and the time-space parameter input with shape $\text{seq} \times C_2$, were normalized and entered into the inEmbed with shape $\text{seq} \times (C_1 + C_2)$. In this work, $C_1 = 14$ represents the number of satellite observations, and $C_2 = 7$ represents the number of input time-space parameters. The inEmbed uses an FNO to encode the input. For a signal $v \in \mathbb{R}^{\text{seq} \times (C_1 + C_2)}$,

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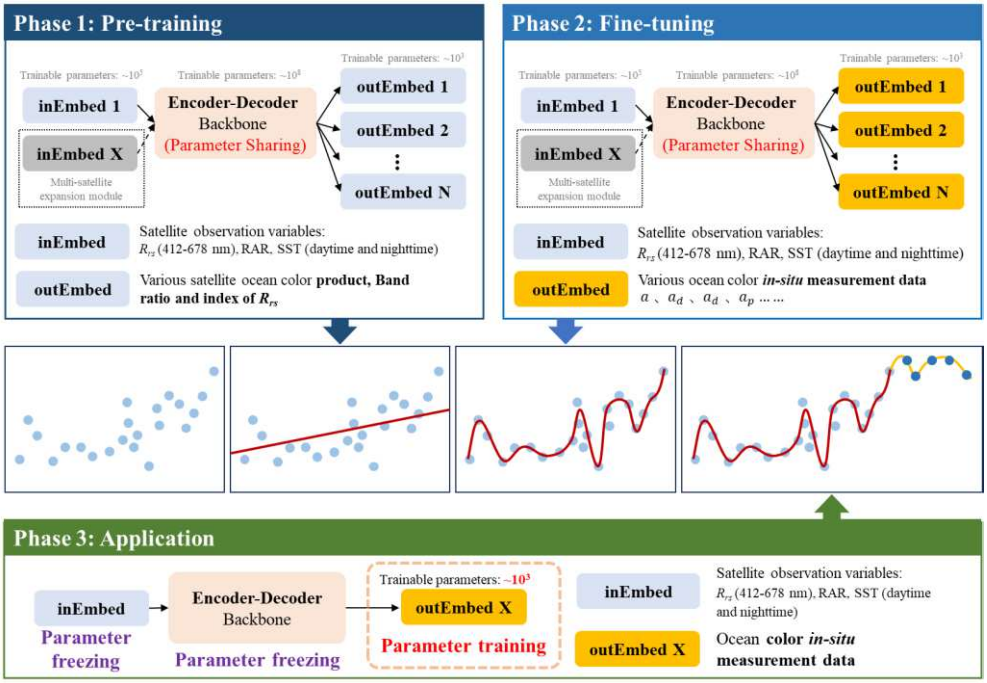


Fig. 5. Three training phases of OCFM: Phase 1—Pretraining with NASA ocean color products; Phase 2—Fine-tuning with in situ measurements; Phase 3—Application for retrieving additional ocean color properties.

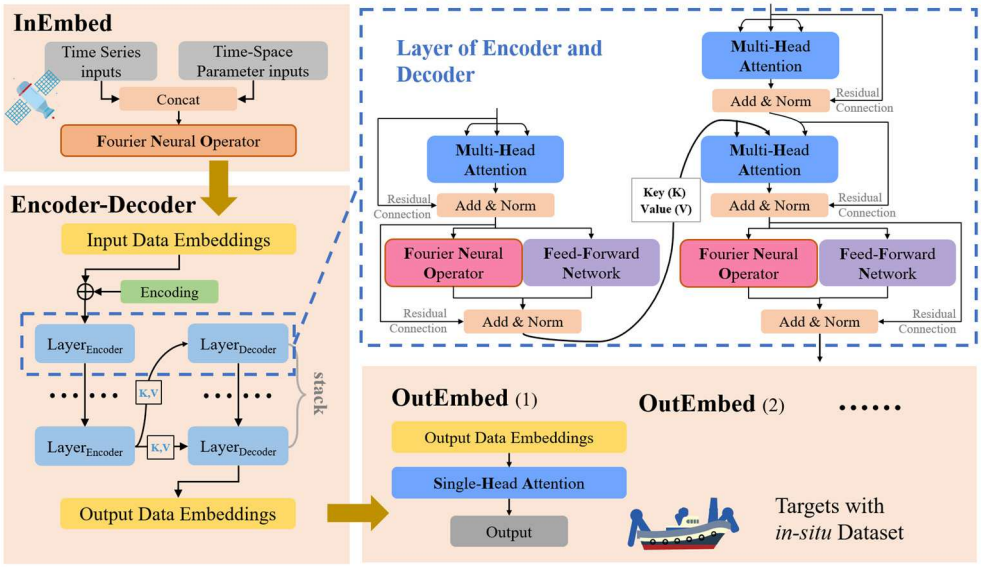


Fig. 6. Framework of OCFM: Satellite data sequences are encoded by the inEmbed module, processed through the encoder–decoder, and then passed through different outEmbed modules to retrieve results for various variables.

processed by the FNO, its representation is explicitly given by

$$\mathrm{FNO}(v) = \mathrm{F}^{-1}\left(R\left(\mathrm{F}(v)_{\mathrm{filtered}}\right)\right) + W(v) \tag{1}$$

where $F(\cdot)$ and $F^{-1}(\cdot)$ denote the Fourier transform and inverse Fourier transform, respectively. The term $R(F(v)_{\mathrm{filtered}})$ represents a linear transformation R applied to the lower Fourier modes while filtering out the higher modes, and W denotes a local linear transformation. This formula has been theoretically and experimentally validated in [61].

After the inEmbed encoding, original inputs form input data embeddings with the shape of $n_{\mathrm{seq}} \times d_{\mathrm{model}}$, where d_{model} represents the combined size of C_1 and C_2 after encoding. The input data embeddings in the encoder–decoder are processed through a sequence encoding mechanism to enhance attention, calculated as follows:

$$\mathrm{SE}_{\mathrm{day}=i} = \cos\left(i\pi/\mathrm{seq}\right). \tag{2}$$

2) OutEmbed: Each outEmbed consists of a single-head attention (SHA) structure. Given the query (Q), key (K), and value (V) matrices produced by the linear transformation of

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the previous layer input, the SHA mechanism is expressed as follows:

$$\mathrm{SHA}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}_{\mathrm{model}}} + B + \mathrm{Mask}\right) \quad (3)$$

where B represents the relative position bias encoding of the input time series, while the Mask is applied to manage missing satellite data. For the satellite sequence with input length seq, the attention matrix B captures the relative positions [62], [63]. Since the possible relative positions range from $[-\mathrm{seq}+1, \mathrm{seq}-1]$, the model includes a smaller learnable positional bias matrix, denoted as $\hat{B}$. The values of B are directly derived from this $\hat{B}$ matrix. The outEmbeds are designed for deriving different ocean color variables from the backbone. Each outEmbed targets a specific retrieval variable, focusing only on the corresponding variable information with the SHA.

After processing by the encoder–decoder, the output data embeddings are obtained as the final abstract features. Afterward, the output data embeddings are entered into the specific outEmbed for the corresponding ocean color variable to obtain the estimation results.

3) Encoder–Decoder: The encoder–decoder backbone serves as the backbone of the model. This backbone integrates the structures of MHA and FNO. The relationship between MHA and SHA is as follows:

$$\mathrm{MHA}(Q,K,V) = \mathrm{Concat}(\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_h)\, W^O \quad (4)$$

$$\mathrm{head}_i = \mathrm{SHA}\left(QW_i^Q, KW_i^K, VW_i^V\right) \quad (5)$$

where W_i^Q, W_i^K, W_i^V, and W^O are neural network learnable weight matrices. For embedded satellite sequences, MHA focuses on the entire sequence information. Concurrently, incorporating FNO into the encoder–decoder backbone enhances the model's reasoning and computational capabilities. Multiple MHA and FNO units form a sublayer, with the encoder and decoder components connected by distinct layers.

The data first passes through a series of N encoder layers in the encoder–decoder backbone. Each encoder layer computes self-attention using an MHA. This is followed by a parallel structure of FNO and feed-forward network (FFN) to process the data. Residual connections are used to maintain performance as the network deepens. The data processed through the N encoder layers is input into the N decoder layers. Key and value weights are computed separately and simultaneously for each decoder layer. In each decoder layer, the data first undergoes self-attention computation with an MHA, as described in (4). It then passes through another MHA to compute attention with the prior key and value inputs, followed by processing through a parallel structure of FNO and FFN. Residual connections are again utilized to maintain performance as the network deepens. After passing through the N decoder layers, the data, now referred to as output data embeddings, leaves the encoder–decoder backbone.

First, the common encoder–decoder allows these small tasks to share sampling information from different geographical areas, effectively linking the inversions of various variables within the model. The backbone network learns the characteristics of all ocean color variables, enabling the retrieval of a specific variable to rely on its own data and the support and constraints provided by other ocean color variables during the training process. Second, each subtask is designed to be small enough to be trained effectively on its limited data, overcoming the limitations imposed by the restricted sampling regions of small datasets.

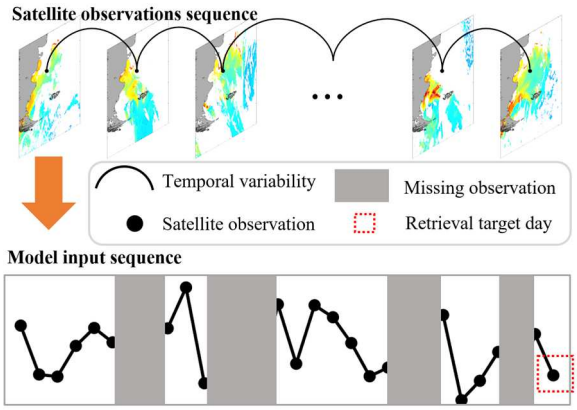


Fig. 7. (Top) Schematic illustration of the satellite observation sequence showing temporal variability, missing observations. (Bottom) Corresponding model input time series with missing data indicated.

OCFM performs gap-filling by utilizing the time sequence from past observations. As illustrated in Fig. 7, even when no valid observations are available for the target day requiring inversion, OCFM can reconstruct the corresponding values based on valid observations from the input time series. To perform gap-filling, the OCFM model leverages a time sequence window of length seq, containing ocean color observations up to the target day m. Formally, the input sequence is defined as

$$\mathbf{x} = \left[x_{m-\mathrm{seq}+1}, \ldots, x_{m-1}, x_m\right] \quad (6)$$

where $x_t \in \mathbb{R}^C$ denotes the multichannel observation at day t, and C is the number of variables. Missing values in the sequence are represented as "NaN," i.e.,

$$x_t = \mathrm{NaN}, \mathrm{if\ the\ observation\ at\ day}\ t\ \mathrm{is\ missing}. \quad (7)$$

Even if the target value x_m is missing, the model can still exploit partially valid information from the past $n_{\mathrm{seq}} - 1$ days. As long as $\exists t \in [x_{m-\mathrm{seq}+1}, m]$, such that $x_t \neq \mathrm{NaN}$, the model can make an estimation.

The OCFM employs an MHA-based architecture, where the self-attention mechanism enables the model to dynamically weigh and integrate available temporal signals. Given the input sequence x, the model learns a mapping

$$\hat{x}_m = \boldsymbol{f}_{\boldsymbol{\theta}}(x) \quad (8)$$

where $\boldsymbol{f}_{\boldsymbol{\theta}}$ denotes the neural network with parameters, $\boldsymbol{\theta}$ and $\hat{x}_m$ are the estimated value at day m. The model is trained to minimize the reconstruction loss over available ground-truth targets, enabling it to infer missing values in the presence of partially observed sequences.

C. Implementation of the Three-Phase OCFM Development

To achieve high-precision and multiple variables retrieval, we divided the OCFM training process into three independent

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phases, as shown in Fig. 5. Each phase with its specific datasets:

1) Phase 1: Pretraining: This phase enables the network to learn the fundamental theoretical relationships embedded in satellite ocean color algorithms developed by earlier generations. This phase utilizes the pretraining database. The input data are MODIS satellite AOP data. The output targets are NASA satellite products derived from MODIS satellites through various algorithms and multiple band ratios, and the NI data calculated from MODIS satellite AOP data. Data before the year 2023 from the pretraining set is used to train the network (including train data points more than 3.6×10^{10}), while the year 2023 data serves as the test set to evaluate the pretraining performance (including test data points about 1.8×10^{9}). Ablation experiments (Section S3 of the Supplementary Material) are also conducted to finalize the network structure. The final model was configured with hyperparameters $d_{\text{model}} = 256, N_s = 6$ and head = 4.

2) Phase 2: Fine-Tuning: This phase is to fine-tune the model based on the most accurate available in situ data–multiple in situ databases. MODIS satellite AOP data serve as the inputs, while ocean color in situ measurements are used as the output targets of the training/validation dataset. This phase refines the model with all collected ocean color variables, including 105 440 sample points, except primary productivity and water clarity. A fivefold leave-one-out cross-validation method is employed to train and validate the network using this in situ database, with the best-performing model retained. Specifically, the dataset was randomly divided into five subsets for each variable type. Each iteration used four subsets for training and one for testing. This process was repeated five times, ensuring that all data points were evaluated once in an independent test. The final performance metrics were aggregated from the results of these five rounds, providing a robust evaluation of the model's fine-tuning effectiveness. The fine-tuning process proceeds as follows: based on the model pretrained in phase 1, we first reduce the learning rate and freeze the parameters of the backbone network. Only the task-specific modules are trained using the in situ dataset during this initial stage. Afterward, all model parameters are unfrozen, and joint fine-tuning is performed on both the backbone and task-specific outEmbed modules. This two-stage strategy enables the model to effectively adapt to high-quality field measurements while preserving the general representations learned during pretraining. Both phases 1 and 2 optimize all model learnable parameters to achieve the most accurate retrieval results.

3) Phase 3: Application: This phase focuses on deployment for end-user extensions. The input and output targets remain the same as in phase 2. End-users can implement ocean color variables retrieval by utilizing the trained OCFM in phase 2 alongside their own smaller datasets. Without the pretraining and fine-tuning in phases 1 and 2, the small datasets from end-users would be insufficient for reliable retrieval. Technically, phase 3 leverages continuous learning through the already trained inEmbed and encoder–decoder backbone. End-users can independently train small subtasks with parameter adjustments limited to the outEmbed, enabling lightweight deployment and efficient adaptation to specific tasks. In situ measurements of primary productivity and water clarity, which were separated in the in situ dataset and not used in phase 2, were inverted as small subtasks to assess the model application potential from the end-user perspective. Water clarity was matched with 517 data points, while primary productivity was matched with 602 data points. The datasets for primary productivity and water clarity were divided based on time. For primary productivity, data collected before 2012 were used for training, comprising 406 sample points, while data from 2012 onward served as the test set, consisting of 111 sample points. For water clarity, the training set included 499 samples collected before 2016, and the test set included 103 samples from 2016 onward.

D. Training Setups of OCFM

The neural network is optimized at each step using the mean squared error (mse) loss function, with additional regularization terms to prevent model overfitting. These regularization terms help the model generalize better by discouraging overly complex models that fit the training data too closely

$$\text{Loss} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \frac{\lambda}{2}\|\mathbf{w}\|_2^2 \tag{9}$$

where N is the number of each batch in each model training, y_i is the target value, $\hat{y}_i$ is the model output value, and $\mathbf{w}$ is the model learnable parameter vectors. The term $\|\mathbf{w}\|_2$ refers to the Euclidean (two-norm) of the vector $\mathbf{w}$, and $\|\mathbf{w}\|_2^2$ represents the square of the Euclidean norm, which is the sum of the squares of the components of $\mathbf{w}$. The regularization coefficient λ, which controls the strength of the regularization term.

Our model is designed to be generalizable, making the optimization process straightforward. We used standard hyperparameters commonly employed in deep learning for the optimization. In phase 1, a learning rate of 0.0002 was used, with a loss function for parameter learning and a regularization term of $\lambda = 10^{-5}$. In phase 2, the learning rate was reduced to 0.0001 while keeping the same regularization term ($\lambda = 5 \times 10^{-6}$). Two phases employed the Adam optimizer [64] and were trained for 200 epochs. These training durations were sufficient to ensure convergence and achieve optimal model performance. We performed additional training with different configurations for the OCFM internal dimension and specific hyperparameter tuning. The results of these ablation experiments are provided in Section S3 of the Supplementary Material.

E. Classification of Water Bodies

Distinguishing between case-1 and case-2 water bodies is crucial for accurately retrieving ocean color properties [65], [66], [67]. These water bodies differ based on the dominant constituents: case-1 waters are dominated by phytoplankton, while case-2 waters are influenced by other particles [66]. In the OCFM, we did not explicitly differentiate the inputs between case-1 and case-2 waters. To demonstrate the model's ability to learn these distinctions from the data automatically,

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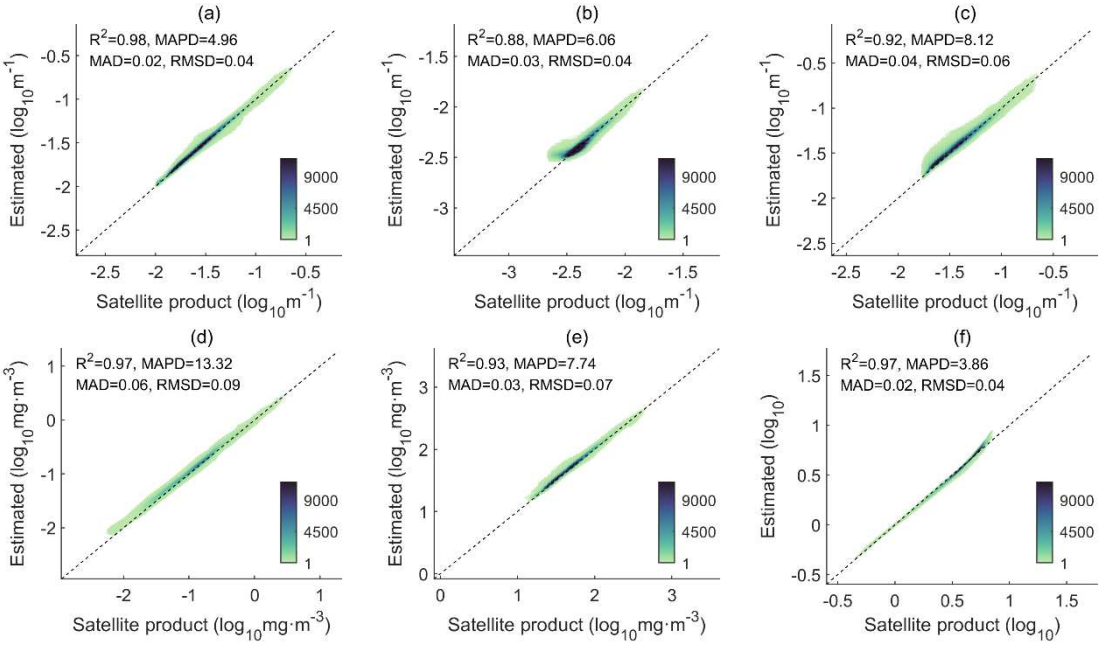


Fig. 8. Comparison of ocean color products and OCFM estimates. (a) Absorption coefficient. (b) backscattering. (c) diffuse attenuation at 490 nm. (d) chlorophyll-a. (e) particulate organic carbon. (f) Ratios between 547 and 443 nm.

we use the standard discrimination criteria from [68] criterion. This criterion employs an inclusive quantitative case-1 standard with parameters $\gamma = 0.1$ and $\nu = 0.5$ to distinguish between case-1 and case-2 waters. Based on this classification, we compare the model's inversion performance for the two water types in subsequent evaluation tasks, in order to examine its adaptability across diverse aquatic environments.

F. Evaluation Metrics

The following evaluation metrics were selected to evaluate the variables estimated by the OCFM:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left[\log_{10}(y_i) - \log_{10}(\hat{y}_i)\right]^2}{\sum_{i=1}^{N}\left[\log_{10}(y_i) - \log_{10}(\bar{y})\right]^2} \tag{10}$$

$$\text{MAD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\text{abs}\left[\log_{10}(y_i) - \log_{10}(\hat{y}_i)\right]} \tag{11}$$

$$\text{RMSD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[\log_{10}(y_i) - \log_{10}(\hat{y}_i)\right]^2} \tag{12}$$

$$\text{MAPD} = \frac{100\%}{N}\sum_{i=1}^{N}\text{abs}\left(\frac{y_i - \hat{y}_i}{y_i}\right) \tag{13}$$

where $\hat{y}_i$ is the model output value, y_i is the target value, and $\bar{y}$ is the mean of the target values. Due to the wide dynamic range of concentration values, which often spans several orders of magnitude, directly computing the coefficient of determination (R^2), mean absolute difference (MAD) and root mean squared difference (RMSD) on the raw data may not be representative. Therefore, we applied a base-10 logarithmic transformation to the raw values before calculating MAD and RMSD, allowing the metrics to reflect model performance across variables better. The R^2 measures the proportion of the variance in the dependent variable that is predictable from the independent variables. The MAD measures the average absolute difference between predicted and actual values, while the RMSD captures the square root of the average squared differences, emphasizing larger differences. The mean absolute percentage difference (MAPD) measures the average absolute percentage difference between the predicted and actual values, providing a relative difference assessment.

IV. Results

The OCFM was developed sequentially in three phases. These three subsections present the results for each phase of the training process.

A. Pretraining Performance in Phase 1

In phase 1, satellite-derived predictors from 2002 to 2022 are utilized as input features to estimate target variables derived from satellite remote sensing products. Importantly, this phase does not incorporate any in situ observations, as both the input and target datasets are exclusively constructed from satellite-based sources. The OCFM learns the theoretical relationships between ocean optical features and R_{rs}, from the ocean color product, band ratio, and NI. It takes 72 h for four NVIDIA A100 80G graphics cards to complete the training. After training, a temporally segmented test set from the year 2023 was utilized to evaluate performance on three critical ocean color variables: $a_{443\text{nm}}$, chlorophyll-a, and $K_{d,490\text{nm}}$, as shown in Fig. 8.

Fig. 8 presents three representative variables from phase 1 (pretraining), where the output targets are satellite-derived products. This phase involves approximately 1.8×10^9 test data points. The goal here is to establish the model's ability to learn general spectral and spatial patterns from large-scale, algorithm-derived datasets. The model takes a 30-day time

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TABLE I

PERFORMANCE METRICS FOR ALL IN SITU DATA USED DURING THE FINE-TUNING PHASE, COMPARED WITH THE CORRESPONDING OCFM ESTIMATES. N DENOTES THE NUMBER OF IN SITU DATA POINTS

symbol	Variables	Unit	N	All				Case-1				Case-2			
				R^2	MAD	RMSD	MAPD	R^2	MAD	RMSD	MAPD	R^2	MAD	RMSD	MAPD
a	absorption coefficient	m^{-1}	5529	0.99	0.03	0.05	6.57	0.99	0.03	0.05	6.44	0.98	0.03	0.04	6.82
a_d	absorption coefficient of detritus	m^{-1}	6786	0.95	0.13	0.18	30.65	0.94	0.12	0.17	29.30	0.93	0.14	0.20	33.46
a_g	absorption coefficient of gelbstoff	m^{-1}	6030	0.97	0.12	0.17	30.54	0.96	0.13	0.18	32.97	0.96	0.11	0.16	25.69
a_p	absorption coefficient of particles	m^{-1}	6757	0.98	0.08	0.10	18.47	0.98	0.08	0.10	18.12	0.97	0.08	0.11	19.17
b_b	backscattering coefficient	m^{-1}	2272	0.97	0.03	0.04	6.36	0.97	0.03	0.04	6.44	0.97	0.03	0.03	5.86
K_d	diffuse attenuation coefficient	m^{-1}	6675	0.94	0.08	0.12	19.26	0.91	0.08	0.12	19.61	0.95	0.08	0.11	17.78
L_w	spectral water-leaving radiance	$\mu W \cdot cm^{-2} \cdot nm^{-1} \cdot sr^{-1}$	9275	0.95	0.10	0.14	24.25	0.95	0.10	0.14	23.21	0.91	0.11	0.15	26.19
E_s	spectral surface irradiance	$\mu W \cdot cm^{-2} \cdot nm^{-1}$	9209	0.86	0.06	0.09	15.47	0.88	0.06	0.09	14.36	0.82	0.07	0.10	17.52
z_1	depth of 1% light level of PAR	m	338	0.88	0.08	0.11	19.77	0.85	0.08	0.11	19.68	0.91	0.08	0.11	20.31
z_{10}	depth of 10% light level of PAR	m	377	0.89	0.08	0.12	19.03	0.84	0.09	0.13	19.66	0.95	0.07	0.09	15.32
z_{37}	depth of 37% light level of PAR	m	387	0.73	0.15	0.21	34.94	0.65	0.15	0.22	36.74	0.83	0.11	0.15	25.38
TChla	total Chlorophyll-a	$mg \cdot m^{-3}$	16595	0.89	0.17	0.24	40.33	0.89	0.15	0.21	35.51	0.79	0.23	0.32	55.05
C_m	micro-phytoplankton	$mg \cdot m^{-3}$	6857	0.93	0.18	0.24	43.76	0.93	0.17	0.23	43.14	0.89	0.20	0.27	47.23
C_n	nano-phytoplankton	$mg \cdot m^{-3}$	7046	0.86	0.16	0.21	38.59	0.87	0.15	0.20	37.58	0.81	0.19	0.26	44.41
C_p	pico-phytoplankton	$mg \cdot m^{-3}$	6888	0.81	0.18	0.26	44.48	0.78	0.18	0.25	42.85	0.87	0.23	0.33	54.22
turbidity	turbidity	NTU	1790	0.50	0.35	0.50	73.35	0.44	0.35	0.49	74.22	0.67	0.35	0.54	62.38
SPM	suspended particulate matter	$mg \cdot L^{-1}$	614	0.55	0.51	0.73	86.29	0.37	0.38	0.53	79.80	0.36	0.65	0.91	94.60
DOC	dissolved organic carbon	$\mu mol \cdot kg^{-1}$	2141	0.66	0.05	0.15	11.80	0.66	0.05	0.15	11.80	0.60	0.06	0.15	11.79
POC	particulate organic carbon	$mg \cdot m^{-3}$	6247	0.48	0.21	0.39	42.09	0.43	0.21	0.39	40.65	0.63	0.25	0.38	57.03
TDN	total dissolved nitrogen	$mmol \cdot m^{-3}$	1400	0.87	0.06	0.09	15.24	0.82	0.06	0.09	15.15	0.92	0.06	0.09	16.15

series (seq = 30) as input and outputs the same value as the standard algorithmic product. To prevent the model from directly relying on the R_{rs} of the final day (day 30) to derive satellite product values, we applied an 80% probability mask to the R_{rs} for day 30. Since band ratios are calculated using data from the last day, masking these R_{rs} values ensure that the model is robust to scenarios where the final day's data is unavailable. This approach enables the model to infer the final day's value based on the preceding sequence. By enforcing this mask setting, the model is trained to learn inversion relationships from the time-series data of days 1 to 29, enhancing its generalization ability and reducing dependency on day 30 data.

The R^2 between the satellite product values and the model estimates exceeds 0.88. The MAPD for the absorption coefficient, backscattering, and diffuse attenuation coefficients are reduced to 4.96%, 6.06%, and 8.12%, respectively, well below 10%. The MAPD for chlorophyll concentration also reaches a relatively low value of 13.32%. In addition, the corresponding MAD and RMSD values remain low across these variables. Most data points are centered around the 1:1 reference line, with only a few outliers distributed farther from the line. The model effectively captures most of the data's inverse relationships in the standard ocean color product. The few outliers are likely caused by the 80% mask. The sparsely distributed outliers correspond to very low densities (low values on the color bar in Fig. 8), indicating that there are few such points overall and that the pretraining results are accurate. A global distribution example of phase 1 is provided in Section S4 of the Supplementary Material.

These satellite ocean color products are MODIS satellite products generated by NASA standard ocean color algorithms, which are widely used [69], [70], [71], [72]. The model's ability to accurately invert the values produced by these algorithms indicates that it has successfully learned the basic inversion relationships from these products. However, there are still discrepancies between the values obtained from these algorithms and the actual ocean environment. Therefore, we use in situ data as the true values for further optimization in the next step.

B. Retrieval of Multiple Ocean Color Properties in Phase 2

1) Accuracy of OCFM Estimates Compared With in Situ Data: In phase 2, the model is optimized using in situ measurement data, which closely represent the real ocean state, with satellite AOP data also serving as inputs. OCFM estimates (outputs) are compared with the in situ measurements (targets of the validation dataset). Each cross-validation process in this phase was conducted on an NVIDIA A100 80G graphics card, requiring approximately 10 h of training. The differences between the estimated results and actual in situ data are evaluated using a fivefold leave-one-out cross-validation method. The inversions of all ocean color properties from phase 2 are presented and analyzed here, with additional analysis focusing on the results for different water bodies. Fig. 9 and Table I correspond to phase 2 (fine-tuning), where the output targets are based on in situ measurements. Because phase 2 uses fivefold cross-validation, the results shown reflect aggregated performance from all validation folds. Figs. 7 and 9 are generated from the same OCFM model at different phases, with varying numbers of active output tasks depending on the task configuration and evaluation focus.

Fig. 9 presents a one-to-one comparison of ocean color variables, with the x-axis representing in situ values and the y-axis showing the corresponding OCFM estimates. For the retrieval of variables, there is a tendency for models to predict the mean value if they are less effective or the dataset is imperfect [73]. Based on the distribution of in situ versus estimated values, there is no indication that the model defaults to mean value estimates. Except for the turbidity

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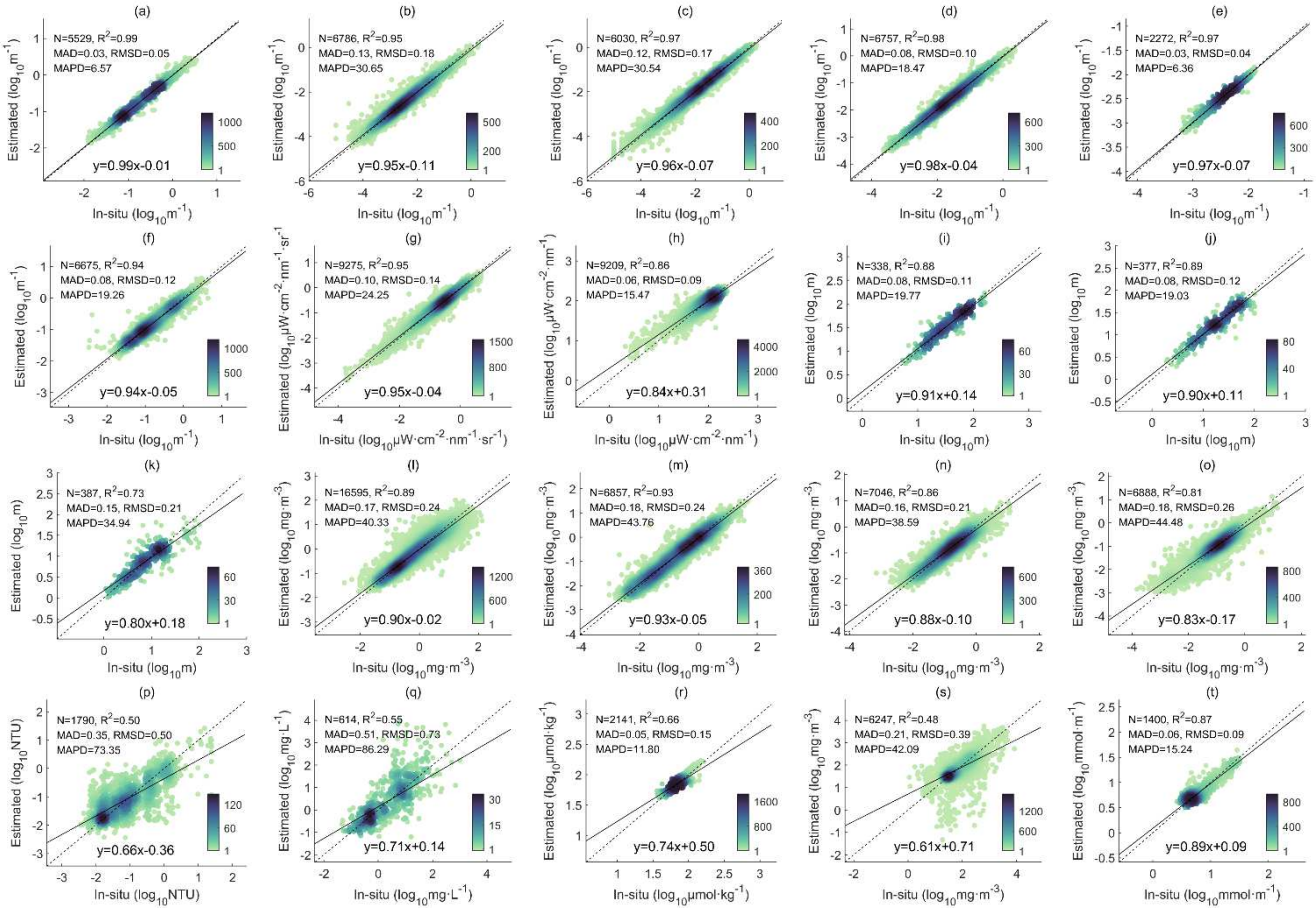


Fig. 9. Comparison of ocean color variables between in situ measurements and OCFM estimated concentrations. (a) a, (b) a_d, (c) a_g, (d) a_p, (e) b_b, (f) K_d, (g) L_w, (h) E_s, (i) z_1, (j) z_{10}, (k) z_{37}, (l) TChla, (m) C_m, (n) C_n, (o) C_p, (p) turbidity, (q) SPM, (r) DOC, (s) POC, and (t) TDN.

regression line, which has a y-intercept of -0.36, likely due to inaccuracies in the in situ data, all other regression lines have very small y-intercepts (less than 0.14 in absolute value). This result indicates minimal systematic difference in the data estimation.

The R^2 and RMSD between in situ measurement and OCFM estimated concentrations are all shown in Table I, where $a, a_d, a_g, a_p, b_b, E_s, L_w$, and K_d contain results from multiple bands. The variables of a, a_d, a_g, and a_p represent different absorption coefficients, achieving R^2 values ranging from 0.95 to 0.99 and MAPD values ranging from 6.57% to 30.65%. Backscattering coefficient b_b achieves an R^2 of 0.97 and an MAPD of 6.36%. K_d achieves R^2 of 0.94 and MAPD of 19.26%. z_1, z_{10} and z_{37} achieve R^2 values ranging from 0.73 to 0.89 and MAPD ranging from 19.03% to 34.94%. This result demonstrates that the OCFM can accurately estimate basic water body optical characteristic variables. For the environmental monitoring factors, the estimated R^2 between the estimated and in situ value for TChla is 0.89, and MAPD is 40.33%. The results of the phytoplankton estimation for the three size classes have R^2 values ranging from 0.81 to 0.93 and MAPD ranging from 38.59% to 44.48%. For SPM and turbidity, as well as other environmental factors, R^2 ranging from 0.48 to 0.87. The results suggest that most of the environmental factors can also be successfully estimated.

The MAD and RMSD for most estimated variables are generally below 0.2 and 0.4, respectively, with the exception of SPM and DOC. Due to the wide dynamic range of these variables, differences are calculated in base-10 logarithmic space to ensure comparability across parameters with differing physical scales. Despite variations in units and magnitudes, the log-transformed differences indicate that model estimates remain close to the corresponding in situ measurements. These results underscore the model's ability to achieve high estimation accuracy across variables spanning several orders of magnitude.

2) Model Performance and Feature Representation Across Different Water Types: For most variables, including absorption coefficients, scattering coefficients, attenuation coefficients, euphotic depth, turbidity, and other environmental variables, the model exhibits comparable inversion accuracy between case-1 and case-2 waters. However, for biologically sensitive parameters such as TChla and PSCs, the accuracy in case-2 waters is notably lower than that in case-1 waters. This outcome aligns with the inherent differences in water composition: case-1 waters are optically simpler, while case-2 waters contain additional optically active constituents, leading to increased inversion complexity. Despite the relatively lower accuracy for case-2 waters, the model still achieves satisfactory performance in these more challenging conditions. To further

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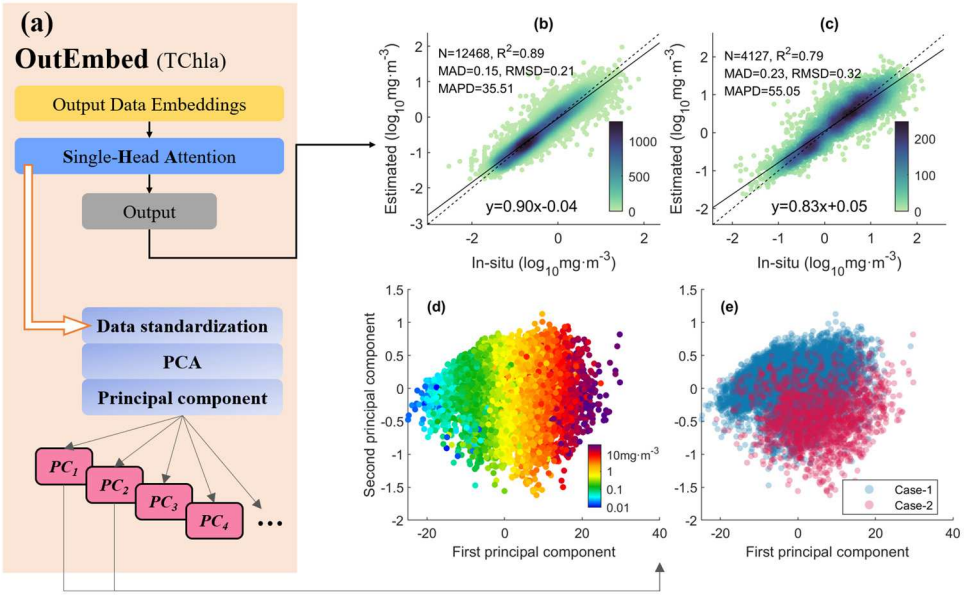


Fig. 10. TChla retrieval results and inner expression of OCFM. The relationship between in situ and estimated TChla concentrations for (b) case-1 water and (c) case-2 water Colormap represents the dot density. PCA decomposition and projection of TChla abstract features. (d) Colored by concentration. (e) Colored by case-1/case-2 water classification.

assess the model's adaptability to different water types, we conducted a separate analysis using TChla as a representative variable. As shown in Fig. 10(b) and (c), the model outputs TChla concentrations for both case-1 and case-2 waters. Case-1 TChla achieved a R^2 of 0.89 with an MAPD of 35.51%, while case-2 TChla had a R^2 of 0.78 and an MAPD of 55.05%. Although the difference in model estimation between the two cases is not substantial, and the data distribution for case-1 and case-2 varies, this result suggests some discrepancy in OCFM estimation of the two water types. Therefore, we further evaluated whether the OCFM can accurately distinguish between case-1 and case-2 water bodies from a deep-learning interpretability perspective.

To achieve this, we extract abstract features from OCFM, which are the data processed in the model's hidden layers and implicitly reflect its potential representation of the data. When the input data enters the model and goes through outEmbed, the values of the hidden layers of the neurons in the SHA are extracted. These hidden layer values serve as abstract features. They are then subjected to principal component analysis (PCA) to obtain the principal components, as shown in Fig. 10(a). PCA is a statistical technique that transforms the original high-dimensional features into a set of orthogonal variables called principal components, which are ranked according to the amount of variance they explain. The first principal component captures the direction of greatest variance in the data, and the second principal component captures the next largest variance orthogonal to the first. These components help to reduce the dimensionality of the data while retaining the most important information. The scores of the two principal component vectors are plotted along the two-axis directions. The first two components account for 99.65% of the total variance, meaning that the two axes in Fig. 10(d) and (e) explain 99.65% of the information with minimal loss. Fig. 10(d) and (e) displays the same point locations but are colored differently to highlight various distinctions. Fig. 10(d) is colored based on TChla concentration, while Fig. 10(e) differentiates between case-1 and case-2 water bodies. From Fig. 10(d), it is evident that there is a strong correlation between the first principal component (PC_1) and TChla concentration. As the PC_1 value increases, the TChla concentration also increases, indicating that PC_1 captures the variations in TChla concentration within the dataset. Fig. 10(e) clearly distinguishes between case-1 and case-2 water bodies along the y-axis (Second principal component, PC_2). The PC_2 values are generally lower for case-2 water bodies, suggesting that PC_2 can capture the differences in optical properties between different water bodies. The results prove that the OCFM includes information relevant to differentiating between case-1 and case-2 waters.

V. Application With Sparse and Few-Shot Data

After completing phase 1 pretraining and phase 2 fine-tuning, phase 3 can be applied. Phase 3 training allows end-users to develop algorithms based on the limited dataset. Most ocean color datasets generally come from data collected during individual research cruises, which are typically limited in scope and geographic range. If these datasets do not encompass the diverse characteristics of marine ecological regions worldwide, the ocean color algorithms that depend upon such datasets cannot be effectively generalized [1], [73]. However, the OCFM backbone network has been trained globally, extracting features from diverse ocean environments worldwide. Consequently, the ocean color retrieval task can be efficiently executed based on the trained model backbone, even when the dataset's geographic coverage is limited. The retrieval accuracy depends on the data provided and the data trained during phases 1 and 2. In this subsection, we retrieve

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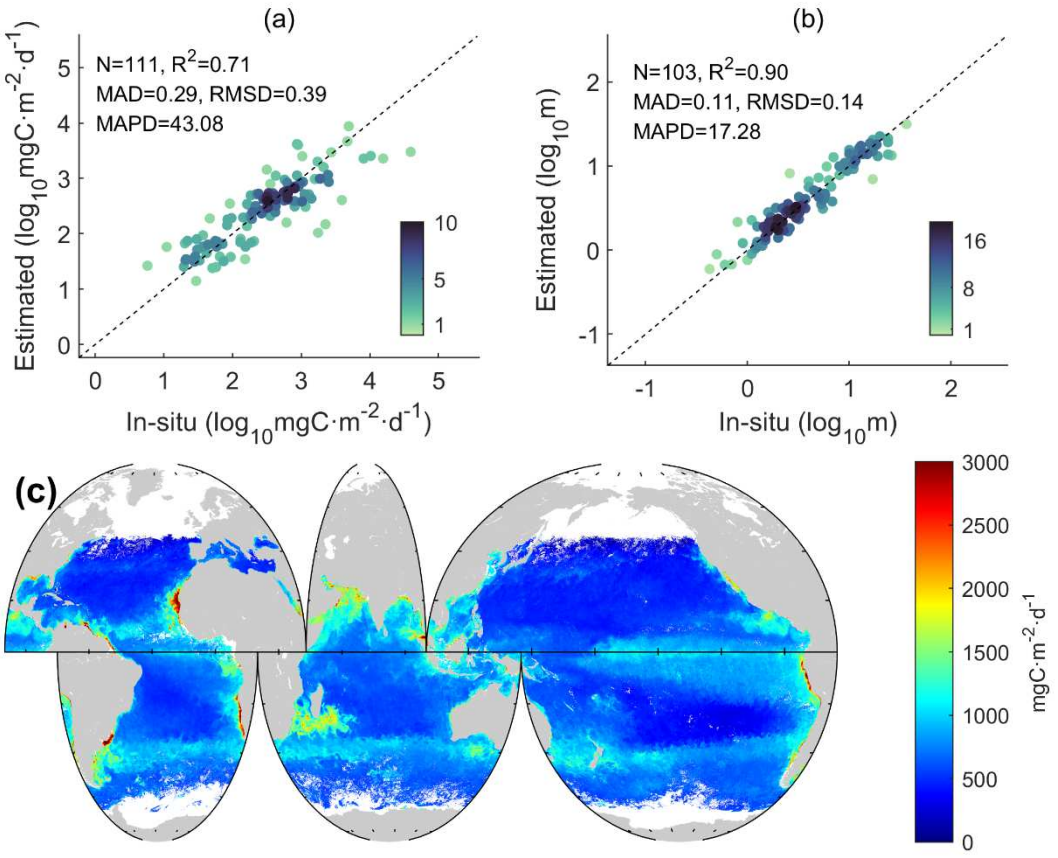


Fig. 11. Application results illustrating. (a) Relationship between in situ and estimated values in the test set for primary productivity. (b) Relationship for water clarity. (c) Example of the global distribution of primary productivity on January 1, 2020.

two variables using OCFM from the user perspective to verify the model application potential. Primary productivity and water clarity are used here as examples, as both are important marine environmental properties [14], [74], [75]. It is worth noting that these two variables are not in the training process of our pretrained model.

We created two new outEmbeds, one for each variable. The training setup followed the same steps as in phase 2, with satellite AOP data as inputs and in situ measurements as target outputs. The test set, comprising data from the year 2012 to 2023 for primary productivity and from the year 2016 to 2023 for water clarity, was relatively small, representing approximately 20% of the total dataset for each variable. Since phase 2 has optimized the model backbone, the learnable training parameters were limited to the outEmbeds. In phase 3, the learning rate was increased to 0.001, with the regularization term remaining at $\lambda = 10^{-5}$. The model was trained quickly on a desktop CPU, completing 200 epochs in 5 min. Point-to-point comparison between the in situ data of the test set and the model output is shown in Fig. 11.

Compared with in situ measurements, the estimated R^2 for primary productivity is 0.71, with an MAPD of 43.08%, while the estimated R^2 for water clarity is 0.90, with an MAPD of 17.28%. Although the R^2 for primary productivity is only 0.71, largely due to the inherent difficulty in estimating primary productivity, which has a high degree of variability [32]. For water clarity, the de-logged RMSD is 0.27 m. Both accuracies are comparable to those of established algorithms at this phase [14], [75], [76]. The development of these algorithms requires significant ocean color expertise, whereas OCFM offers a readily available, mature solution. Leveraging the well-trained models from the first two phases, high accuracy can be achieved in this phase even with limited data and computational resources.

Global satellite data can be input into the trained model to generate and analyze ocean color property distributions, as shown in the OCFM-derived primary productivity map on January 1, 2021 [see Fig. 11(c)]. The data entered into the model is a time-series sequence with a length of seq = 30. If there are no observations for day =30, the model can still estimate day =30 using observations from prior days. Primary productivity is concentrated in mid-to-high latitudes, nearshore areas, and upwelling zones, ranging from below 500 $\text{mgC} \cdot \text{m}^{-2} \cdot \text{d}^{-1}$ in the open ocean to approximately 1500 $\text{mgC} \cdot \text{m}^{-2} \cdot d^{-1}$ in productive regions, peaking at 3000 $\text{mgC} \cdot \text{m}^{-2} \cdot d^{-1}$ in the highly productive zones. These features closely align with findings from previous studies [77], [78], [79].

Ocean color satellite data often has missing values due to clouds, orbital coverage, and other issues. However, its time-series information captures rich ocean color features, reflecting temporal and environmental changes [58], [80], [81], [82]. Using the time series as model input, OCFM can still perform

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ocean color retrieval tasks even when data for a specific day is missing, as long as data from other times in the series are available. Many ecological models, such as the NPZ and NPZD models [57], [83], are based on time-varying equations. OCFM operates in a manner similar to ecological modeling; however, its inverse relationships, derived from data via deep learning, offer greater flexibility and accuracy compared with the fixed, prior knowledge typically employed in ecological models. The data interpolating empirical orthogonal functions (DINEOFs) [84], [85] data relies on the singular value decomposition, with its accuracy dependent on the algorithm of the corresponding interpolated satellite product [86]. OCFM uses satellite products for pretraining and in situ measurements for fine-tuning, offering higher data accuracy compared with DINEOF. However, OCFM cannot achieve the complete global coverage that DINEOF provides for extensively missing observations in time series. Overall, OCFM still has some ability to generate complete global distribution data, illustrating the potential for OCFM applications.

VI. Discussion

A. Necessity of the Three Training Phases

Although we collected as much in situ measurement data as possible ($\sim 10^5$), this number is still far less than the number of inner learnable parameters in the model ($\sim 10^8$). Fitting a model with more parameters than observations can be problematic [87], making the pretraining step in phase 1 essential. Each satellite-derived product pixel in the daily images served as a training data point. This resulted in over 3.8×10^{10} training samples, providing a large-scale dataset to support robust pretraining of the OCFM. While the algorithmic product data differs from real-world results, it represents various theoretical aspects of ocean color algorithms. Learning from this data transforms the model from an initially disordered state to one adapted for ocean color inversion.

Phase 2 involves fine-tuning the model from phase 1. Since the data in phase 2 is derived from global in situ measurements, which closely approximate real-world values, this fine-tuning shifts the model focus from theoretical ocean color knowledge to real-world ocean conditions. In addition, by simultaneously learning multiple ocean color variables, phase 2 allows the model to understand the potential equilibrium and constraints among these ocean color properties. These variables are difficult to represent in a single ocean color algorithm [17]. Once fine-tuned across a comprehensive range of ocean variables, the model becomes a generalized foundation model capable of inverting all ocean color properties. This foundation model is the base upon which almost all ocean color properties can be inverted through further training or fine-tuning, such as phase 3.

For global observations, an important indicator of ocean color algorithms is their ability to be deployed and applied lightweight manner. The core objective of retrieving various IOPs from AOPs is to model light transport through the water column [88], [89]. This process relies on the radiative transfer equation and other fundamental laws of physics [89], [90], [91]. However, these simulations require considerable computation and are challenging to apply on a large scale. Therefore, various analytical and semi-analytical algorithms have been developed for quick and efficient application [6]. Large deep-learning models with numerous parameters are difficult to deploy. Our model, with a parameter count of 10^7–10^8, demands significant computational resources during training. However, once trained, OCFM can efficiently perform inference on a standard desktop CPU. In the phase 3 application, accurate retrieval was achieved with training only on outEmbed for a new ocean color variable. The number of new outEmbed training parameters ranges from 10^3–10^4, making it a very small learning task. Therefore, the future deployment and application of the trained OCFM can be efficiently applied on a desktop CPU without the need for professional-level supercomputers or advanced graphics cards.

B. Distinctive Features of OCFM

The foundation model is characterized by its generality and adaptability. Unlike empirical algorithms, OCFM relies on extensive prior data during training, resulting in greater universality and robustness. Compared with traditional deep-learning models, OCFM is a generalizable model trained on large-scale data to learn transferable spectral–spatial representations. Once pretrained, it requires minimal parameter updates to adapt to new retrieval tasks. Unlike SeaDAS or the GIOP framework, which often require reconfiguration for each variable or region [25], OCFM is entirely data-driven and learns mappings directly from observations. This process enables consistent and scalable performance across diverse conditions. Fine-tuning involves only lightweight updates, ensuring temporal and spatial consistency without reconstructing the full model.

Most ocean color datasets are derived from data collected during individual cruises, which are typically limited in size and geographic coverage. If these datasets do not capture the characteristics of diverse marine ecoregions globally, the conventional theory results in algorithms that are limited to regional applications, restricting their ability to generalize effectively to other regions or on a global scale [1], [73]. However, the backbone network of our model has been trained on a global scale, extracting features from diverse environments worldwide. Consequently, the global ocean color retrieval tasks can be efficiently executed based on the trained model backbone, even when the dataset's geographic coverage is limited.

The model is inherently extendable to accommodate new satellite instruments. We have designed the inEmbed module with additional input channels, allowing for straightforward integration of data from other satellite platforms. For instance, Lee et al. [31] introduced a deep-learning-based cross-satellite atmospheric correction method, which aligns observations from various sensors to the MODIS standard. This method can function as a preprocessing front-end for our framework, harmonizing heterogeneous satellite data before it is fed into the model. Once calibrated, the new data can be incorporated via an updated inEmbed, enabling seamless adaptation to additional satellite missions without the need for retraining the entire model.

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TABLE II
COMPARISON OF PERFORMANCE METRICS FOR DIFFERENT BACKBONE MODELS, COMPARING IN SITU VALUES WITH CORRESPONDING MODEL ESTIMATES

symbol	RNN				LSTM				Original Transformer				Our			
	R^2	MAD	RMSD	MAPD	R^2	MAD	RMSD	MAPD	R^2	MAD	RMSD	MAPD	R^2	MAD	RMSD	MAPD
a	0.97	0.04	0.07	10.29	0.98	0.03	0.05	7.97	0.97	0.05	0.07	10.93	0.99	0.03	0.05	6.57
b_b	0.90	0.06	0.08	14.61	0.94	0.04	0.06	10.55	0.91	0.06	0.08	13.54	0.97	0.03	0.04	6.36
K_d	0.90	0.11	0.16	26.90	0.91	0.10	0.14	23.18	0.90	0.11	0.15	25.50	0.94	0.08	0.12	19.26
z_1	0.84	0.10	0.13	24.65	0.86	0.09	0.12	22.62	0.85	0.10	0.13	23.45	0.88	0.08	0.11	19.77
TChla	0.78	0.25	0.33	62.83	0.78	0.25	0.34	61.93	0.77	0.26	0.34	64.58	0.89	0.17	0.24	40.33
turbidity	0.38	0.43	0.55	87.03	0.41	0.40	0.54	81.24	0.36	0.44	0.56	87.75	0.50	0.35	0.50	73.35
SPM	0.39	0.61	0.85	90.58	0.52	0.54	0.75	95.33	0.40	0.61	0.84	91.13	0.55	0.51	0.73	86.29
DOC	0.35	0.11	0.20	23.36	0.34	0.11	0.21	25.47	0.29	0.12	0.21	24.80	0.66	0.05	0.15	11.80
POC	0.32	0.25	0.45	44.35	0.38	0.26	0.43	48.19	0.32	0.26	0.45	49.45	0.48	0.21	0.39	42.09
TDN	0.82	0.08	0.10	17.50	0.83	0.08	0.10	17.40	0.81	0.08	0.11	18.58	0.87	0.06	0.09	15.24

Our model is built upon MODIS level-3 products, which provide temporally composited and spatially gridded data at a global scale. These products offer consistent spatial coverage and a fixed grid structure over time, allowing us to construct regular time series at each pixel. This consistency is essential for capturing the spatiotemporal dynamics of ocean color and enables the model to learn temporal dependencies across space and time effectively.

In contrast, MODIS 1-km level-2 products are based on individual satellite overpasses and provide only partial, nonuniform spatial coverage at each time point. The spatial locations observed vary from day to day, making it difficult to align observations into a consistent pixel-level time series required by our model. While it is technically possible to manually assemble level-2 data over a specific region into a custom time-series dataset, such an effort involves considerable preprocessing and is sensitive to data sparsity caused by cloud cover or acquisition gaps. Moreover, unlike level-3 products, which allow direct and efficient extraction of pixel-wise time series, level-2 data lack standardization across time, limiting their suitability for large-scale, automated applications. Therefore, level-3 gridded products remain the preferred input source for our framework.

C. Accuracy and Potential Limitations of OCFM

The analysis shows that OCFM achieves efficiency in three aspects. First, for computational efficiency, phase 3 fine-tuning requires less than five minutes on a CPU and updates fewer than one percent of the model parameters, reducing user resource requirements. Second, regarding data efficiency, the model leverages large volumes of algorithm-derived satellite products during pretraining to learn spectral and spatial representations, reducing dependence on in situ measurements during fine-tuning. Third, OCFM supports multiple retrieval tasks, such as water clarity and primary productivity, for operational efficiency without requiring case-specific model adjustments, unlike traditional methods that depend on manual tuning.

To benchmark our model's performance, we substituted its architecture with several widely used deep-learning models, applying identical training and inversion procedures. Ten variables were selected as representative variables for comparison. The results are summarized in Table II.

When comparing multiple evaluation metrics, our model consistently outperforms the original Transformer architecture, highlighting the performance improvements achieved by adding the FNO structure and positional encoding. In addition, the accuracy of our model surpasses that of other baselines, including RNN and LSTM models. Our model still achieves the highest accuracy among all compared methods in the same dataset.

We also compare OCFM standard products (see Figs. S5 and S6). While OCFM outperforms standard remote sensing products across multiple variables, the baseline methods used for comparison are operationally stable and widely adopted, but they may not incorporate recent advances in variable-specific algorithms. Some modern approaches, particularly those based on refined data from sensors such as Sentinel or GOCI, may yield higher accuracy for individual variables. Nevertheless, OCFM provides robust multivariable performance.

However, turbidity exhibits reduced accuracy compared with other variables, which can be attributed to two main factors. First, there are inherent limitations in the input spectral bands. The MODIS L3 gridded data used in this study do not include near-infrared bands, which are critical for retrieving turbidity, SPM, and similar variables [18], [92]. The absence of these bands likely constrained further improvements in accuracy. Second, the 4-km spatial resolution also influences the retrieval of turbidity. In our preprocessing, spatial matching was performed using a 3×3 pixel window at 4-km resolution, combined with 30-day temporal sequences. While this ensured consistent spatial–temporal alignment, it inevitably smoothed short-term and small-scale variability, thereby limiting the model's sensitivity to rapid changes in highly dynamic variables such as turbidity and SPM, both of which exhibit strong temporal and spatial variability [7], [8], [93], [94]. Although OCFM outperforms existing products overall, turbidity remains a particularly challenging variable.

It is worth noting that OCFM is a foundation model rather than a fixed, task-specific architecture. Its inherent adaptability enables fine-tuning for diverse application scenarios. Future enhancements could be achieved by integrating additional spectral bands and leveraging higher-frequency, higher-resolution satellite observations from platforms such as PACE, GOCI, and Sentinel, as well as incorporating region-specific in situ datasets. In particular, end-users can utilize local measurements tailored to their region or application to

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enable targeted model customization. Such user-driven fine-tuning has the potential to significantly improve retrieval accuracy, enhance responsiveness to dynamic changes, and reduce uncertainty in operational deployments.

D. Potential Application for Different Water Types

The accuracy of TChla in case-1 and case-2 waters shows slight differences, consistent with previous findings [65], [66], [67], [95]. By analyzing the abstract features in TChla outEmbed, the first two principal components of the PCA present clear and consistent results with existing knowledge. The subtask aims to estimate corresponding variable values, so the first principal component is expected to be related to concentration. The second principal component is differentiated across different water bodies. Fig. 11(e) further shows that case-2 water samples include some instances with low chlorophyll concentration, indicating that high TChla values are not exclusive to case-2 waters. This observation is also supported by prior findings [68].

These results suggest that OCFM can learn high-dimensional representations that encode complex water body characteristics, enabling the model to make accurate estimations without relying on explicitly defined categories. Notably, the model is trained using only observational data inputs, without any explicit physical or classification-based priors. The resulting latent water-type representations are learned autonomously, reflecting an emergent property commonly observed in foundation models [96]. This result implies that OCFM shares structural characteristics with foundational architectures, rather than functioning as a conventional task-specific deep-learning model.

While the model demonstrates adaptability to a range of water types, its generalizability has certain limitations. For instance, the 4-km resolution of MODIS level-3 data used in this study is insufficient for resolving the fine-scale spatial heterogeneity of inland waters. Consequently, the current model is focused on marine environments and is not directly optimized for inland water applications, where factors such as optical shallow-water effects and land adjacency significantly influence spectral signals. Nevertheless, the model architecture includes extensible components, such as the inEmbed interface illustrated in Fig. 6, which can be leveraged in future work to accommodate higher-resolution data and a broader range of sensor inputs, ultimately supporting more generalized aquatic inversion tasks.

VII. Conclusion

This article presents the development of a deep-learning foundation model for ocean color properties retrieval. The model trained on ocean color theories and existing operational products, fine-tuned with extensive global in situ data to achieve high precision in retrieving multiple ocean color properties. In its application phase, the model exhibits significant adaptability, enabling the creation of high accuracy, task-specific algorithms even with limited user-provided datasets.

OCFM demonstrates strong generalization capabilities, achieving state-of-the-art performance across multiple ocean color variables with limited supervision. Compared with various models, it demonstrated the best accuracy. The model also maintains physical consistency, with internal feature representations aligning with known oceanographic phenomena such as the distinction between case-1 and case-2 waters. This implicit learning of physical structures is achieved without explicit parameterization, offering a data-driven alternative to traditional semi-analytical approaches. Furthermore, OCFM's modular architecture enables seamless integration of additional variables, such as SPM and primary productivity, thereby overcoming the limitations of case-specific algorithm tuning and enhancing its practical scalability.

By transitioning from fragmented, task-specific algorithms to a unified and adaptable framework, OCFM represents a paradigm shift in ocean color remote sensing. Its ability to generalize across variables, reduce data dependence, and scale with minimal tuning offers a practical and theoretically grounded solution for advancing marine biogeochemical monitoring.

In the future, deep learning in ocean remote sensing is expected to continue expanding and evolving. Unlike task-specific models, OCFM overcomes the need for large datasets and intensive hardware resources, providing a lightweight and generalized solution for few-shot data. Ongoing advancements in foundational model research across various fields have shown groundbreaking progress [29], [30], [97], [98]. We are confident in OCFM's potential for ocean remote sensing, especially in ocean color analysis. Future research directions for OCFM include enhancing its integration with multiple satellite systems to enable more comprehensive monitoring of ocean color properties and specific geographic features, further supporting global marine environment monitoring efforts.

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The OCFM code repository will be available at GitHub: https://github.com/yangyiiocas/Ocean-Color-Foundation-Model. The authors would like to thank the following data sources for providing the datasets used in this study: Daily 4-km level-3 MODIS-Aqua satellite data and products from the Ocean Biology Processing Group (OBPG), available at https://oceancolor.gsfc.nasa.gov/l3/; SeaBASS in situ measurement data from https://seabass.gsfc.nasa.gov/; NOMAD in situ measurement data from https://seabass.gsfc.nasa.gov/wiki/NOMAD; MAREDAT in situ measurement data from https://doi.org/10.5194/essd-5-227-2013; GLODAPv2 in situ measurement data from https://glodap.info/; and other in situ measurement data from https://www.pangaea.de/

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